

Unit 9, Lesson 3: Translations

Benchmarks

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

Learning Targets

- ☐ I can describe relationships between figures after a translation.
- ☐ I can perform a translation of a figure.
- ☐ I can perform a translation of a figure on the coordinate plane.

9.3.1 Warm-Up: Describing Patterns

A pattern is formed using ceramic tiles.



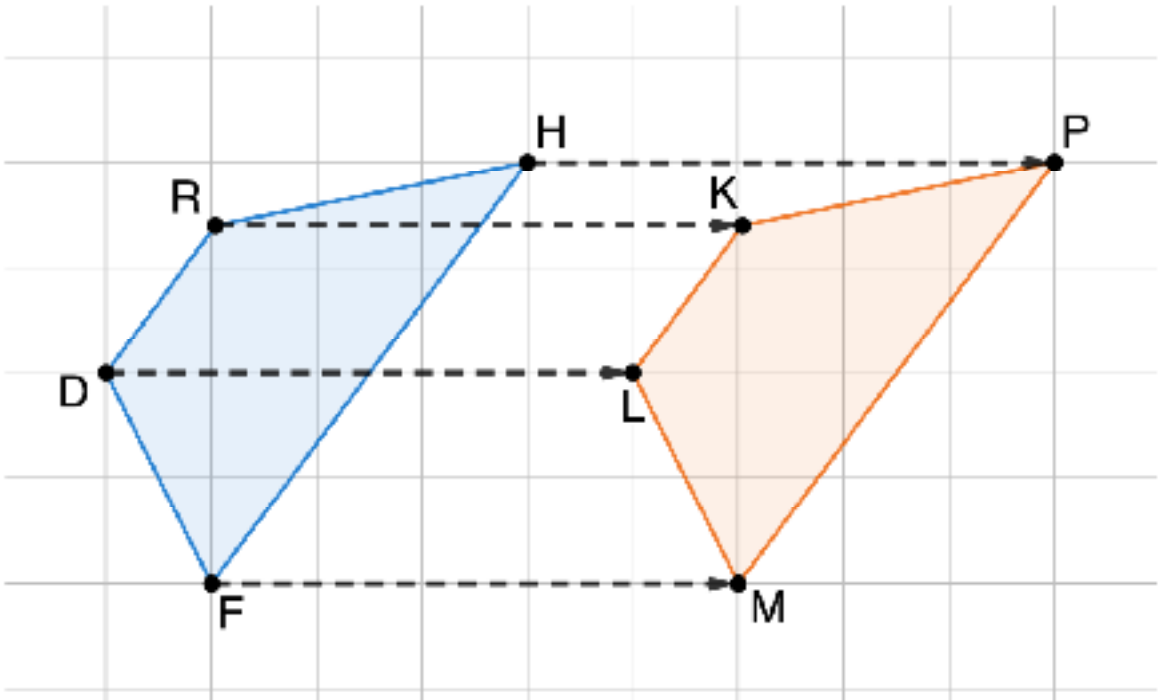
1. Describe the geometric transformations that could have been used to make the pattern.

9.3.2 Exploration: Defining Translations

Work with your partner to complete the following.

The rigid transformation of “sliding” a figure is called a *translation*.

1. In the figure shown, quadrilateral $RDFH$ was translated to create quadrilateral $KLMP$.



- a. Each gridline in the image represents 1 centimeter (cm). Complete the table.

Corresponding Vertices	Distance (cm)
R and K	
D and L	
F and M	
H and P	

- b. What do you notice about the distances from the vertices of the preimage to the vertices of the image?

c. Complete the statements.

Quadrilateral $RDFH$ was translated ____ cm to the

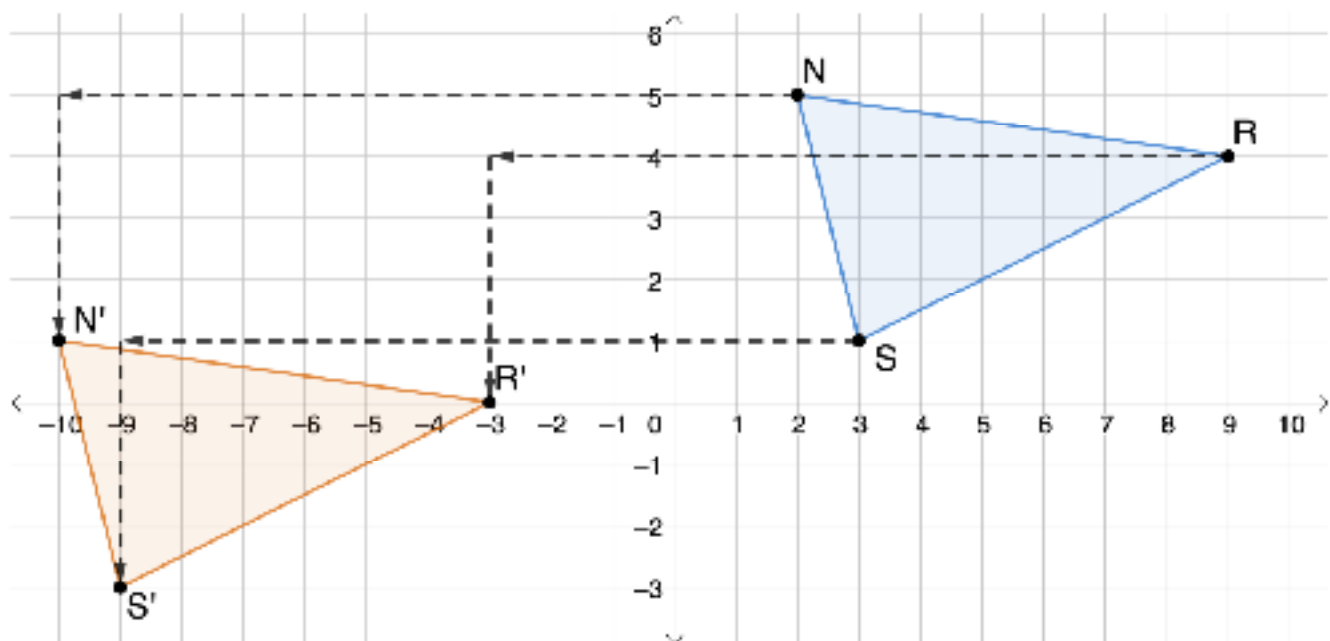
- ☐ left
- ☐ right

to create quadrilateral $KLMP$. The mapping

lines for the translation are

- ☐ intersecting.
- ☐ parallel.
- ☐ perpendicular.

2. Rigid transformations can also be performed on a coordinate plane. The example shows a translation performed on $\triangle NRS$.



Triangle $N'R'S'$ is read as "n prime, r prime, s prime." This notation is often used when a figure has been transformed. When prime notation is used, it is most commonly used for the vertices of the image.

The arrows show a possible path of movement. When a figure is translated on the coordinate plane, the figure's movement is best described as a combination of horizontal and vertical movements, instead of diagonal.

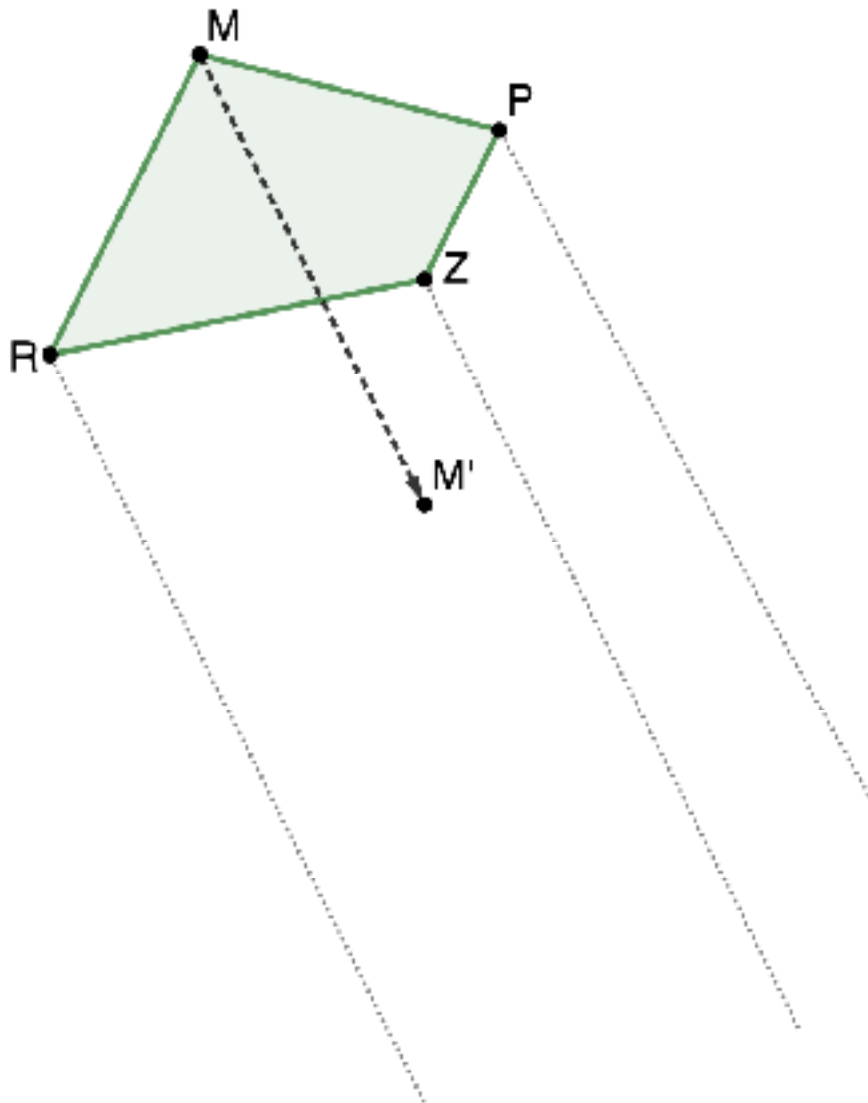
- a. Complete the statements to describe the translation of $\triangle NRS$ to $\triangle N'R'S'$.
- Point N was translated ____ units left and ____ units down to point N' .
 - Point R was translated ____ units left and ____ units down to point R' .
 - Point S was translated ____ units left and ____ units down to point S' .
- b. What do you notice about the number of units and the direction that each point was translated?

9.3.3 Guided Instruction: Translations On and Off the Coordinate Plane



A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

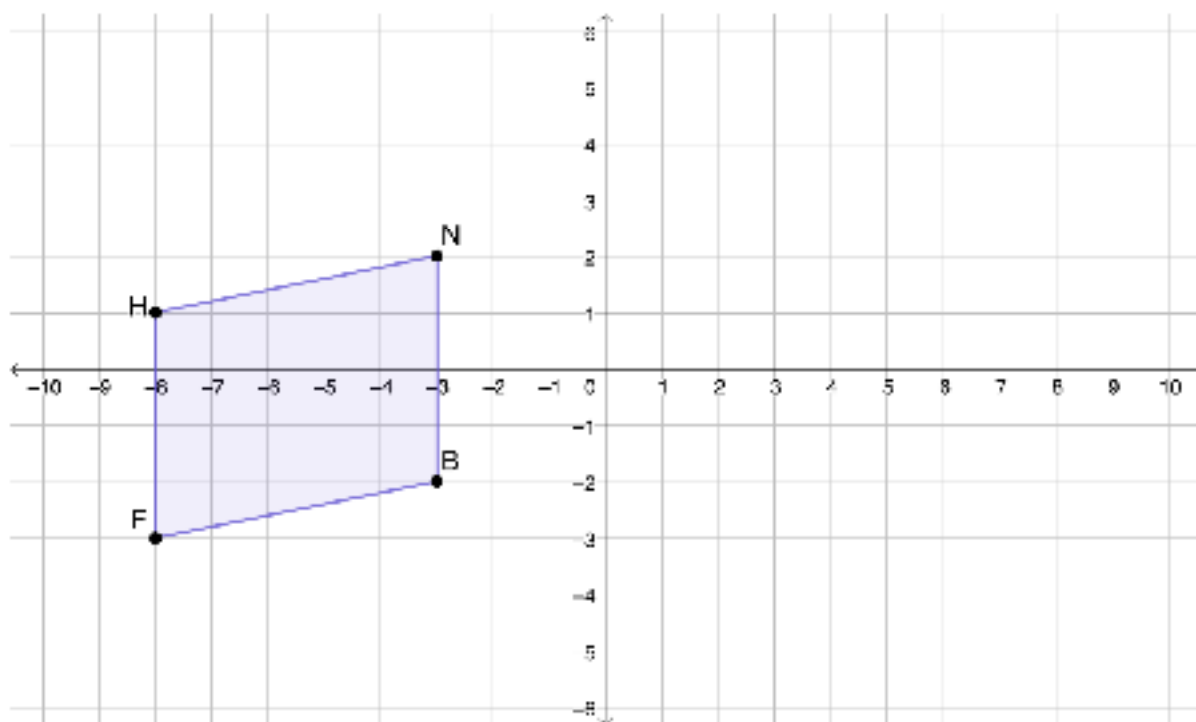
1. The figure shows quadrilateral $MRZP$ and the mapping for vertex M . Assume the transformation is a translation on the plane.



- a. Measure the length of the mapping from point M to point M' in cm. What is the distance?

- b. Parallel line segments that can be used for the mapping are shown on the image above in grey. Use the measurement from Part A and the parallel line segments provided to complete the translation. Draw the mapping from vertices P , Z , and R that are the same length and parallel to the first mapping from point M to point M' . Label the image quadrilateral $M'R'Z'P'$.

2. Quadrilateral $BFFN$ is plotted on the coordinate plane.



- a. Translate quadrilateral $BFFN$ 4 units to the right and 3 units up. Label the image $B'F'H'N'$. Draw the mapping to show how each vertex moves from the preimage to the image.

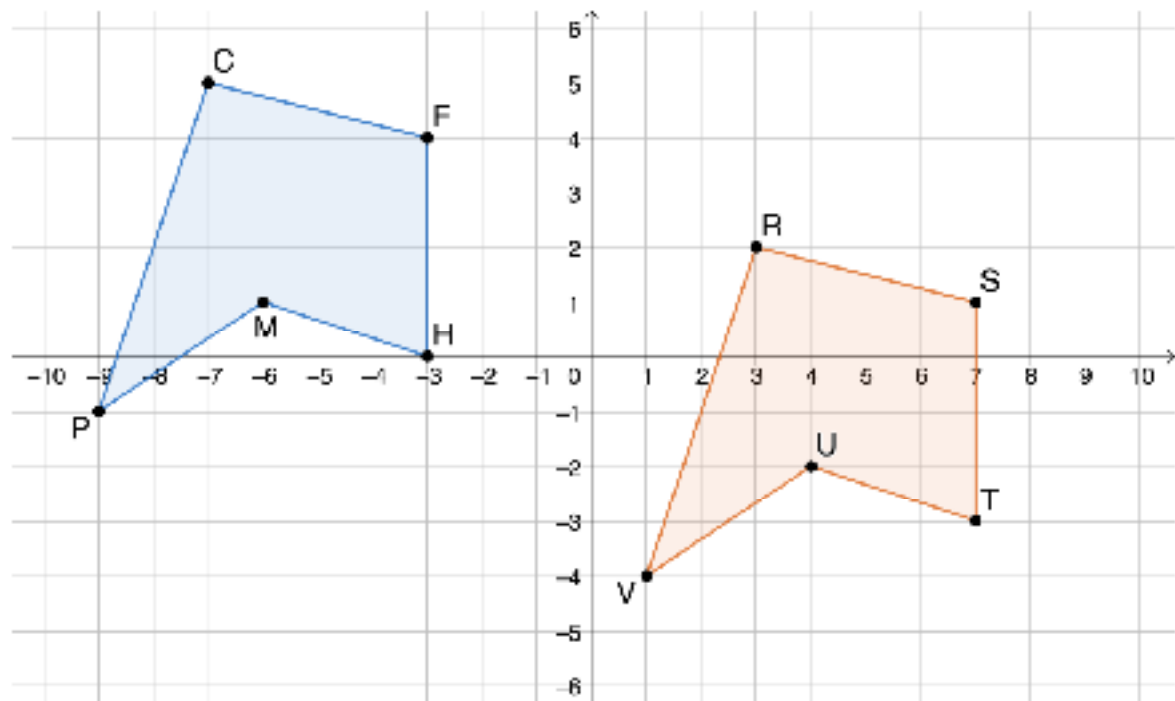
b. Record the coordinates of the vertices of the preimage and image.

	Coordinates		Coordinates
Point <i>B</i>		Point <i>B'</i>	
Point <i>F</i>		Point <i>F'</i>	
Point <i>H</i>		Point <i>H'</i>	
Point <i>N</i>		Point <i>N'</i>	

c. Describe the relationship between the coordinates of the preimage and image.

9.3.4 Try It: Translations On and Off the Coordinate Plane

1. In the figure, polygon *RSTUV* is the result of a translation performed on polygon *CFHMP*.



- a. Record the coordinates of the preimage and the image.

Vertex	Coordinates	Vertex	Coordinates
Point <i>C</i>		Point <i>R</i>	
Point <i>F</i>		Point <i>S</i>	
Point <i>H</i>		Point <i>T</i>	
Point <i>M</i>		Point <i>U</i>	
Point <i>P</i>		Point <i>V</i>	

- b. Complete the statement.

Polygon *CFHMP* was translated _____ units

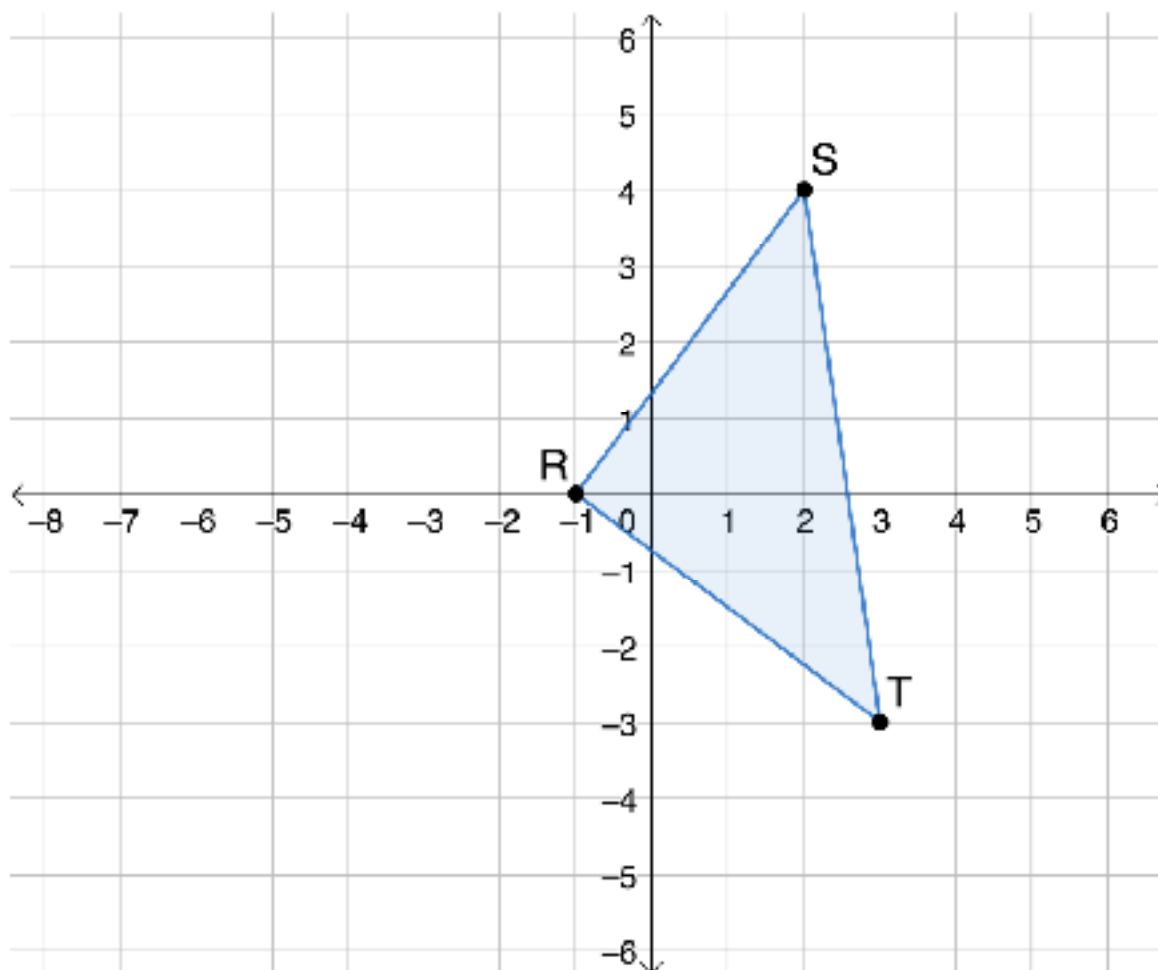
☐ right
☐ left

and _____ units

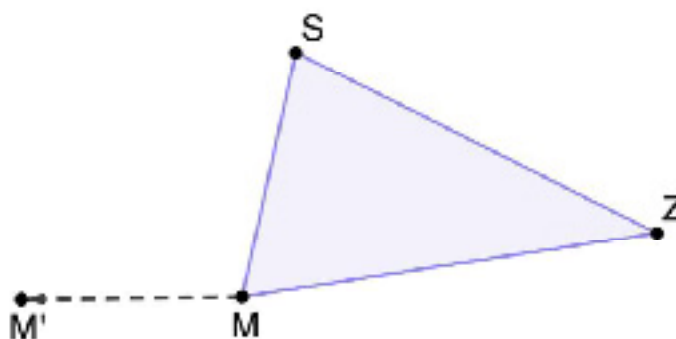
☐ up
☐ down

to create polygon *RSTUV*.

2. Translate $\triangle RST$ 5 units left and 2 units down to form $\triangle R'S'T'$. Label the image and draw the mapping.

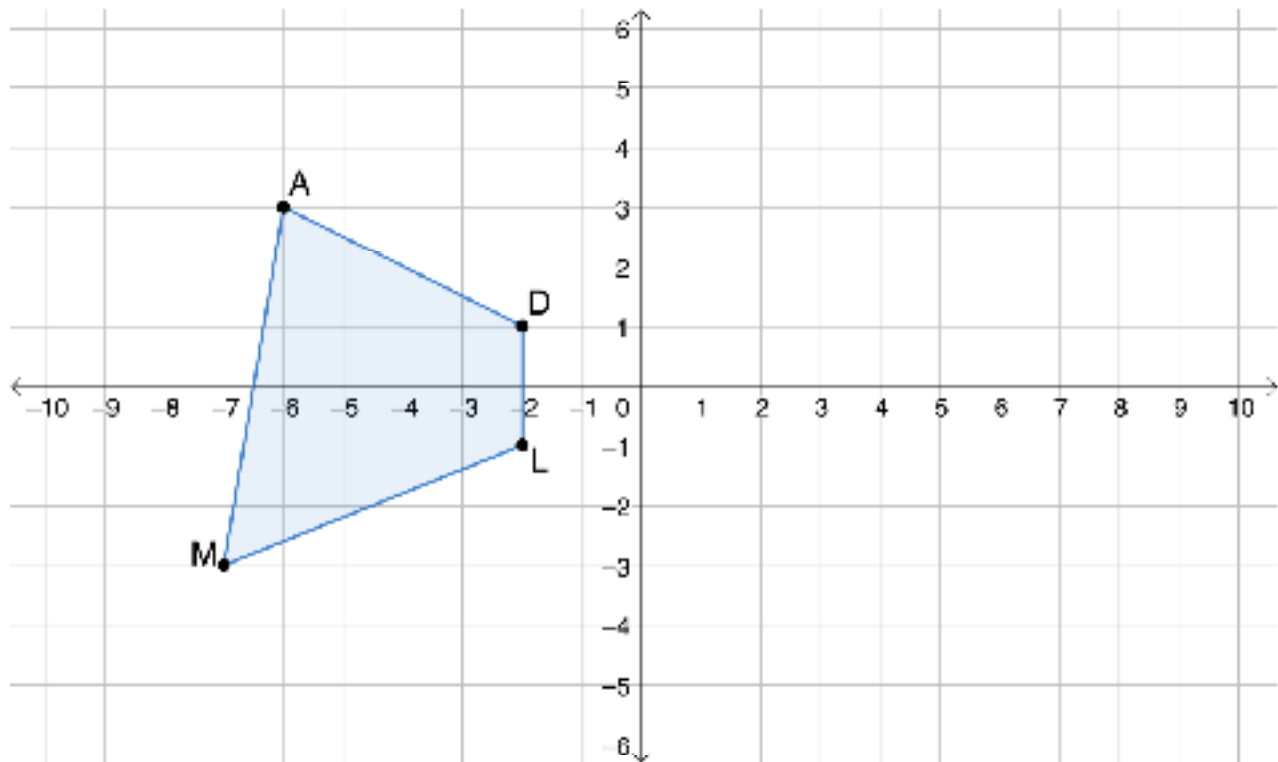


3. Use a straightedge to complete the translation of $\triangle MSZ$ started below. Label the image $\triangle M'S'Z'$, and draw the mapping of vertices S and Z .



9.3.5 Wrap-Up: Ticket Out the Door

1. **Quadrilateral $ADLM$ is shown on the coordinate grid.**
 - a. **Translate quadrilateral $ADLM$ 4 units right and 1 unit up to create the image quadrilateral $A'D'L'M'$. Label the image.**



- b. **Describe at least two characteristics of the relationship between the two figures after the translation.**

